جامعة الفيصل
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College of Engineering and Advanced Computing

Introduction to Artificial Intelligence (SE 444)

Lecture 7.1: Knowledge-Based Agents

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This lecture introduces Knowledge-Based Agents as a fundamental concept in Artificial Intelligence. You'll learn how AI agents utilize formal logic to represent knowledge about their environment and reason to make informed decisions. We will explore key techniques such as **knowledge representation**, **inference mechanisms**, and how agents leverage these for effective **problem-solving**, equipping you with the skills to understand intelligent reasoning systems.



Knowledge-Based Agents: Core Architecture

🧠 Knowledge-Based Approach

- 🧩 **Core Idea:** Agents use a **Knowledge Base (KB)** to **represent** and **reason** about the world.
- 🧱 Maintain **internal models** of their environment for informed decision-making.
- ⚡ **Unlike reactive agents**, they don't just react – they **reason** before acting.
- 🎯 Accept **explicitly defined goals** and plan actions to achieve them.
- 🧠 Gain competence by:
  - 📘 **Being told** new knowledge (declarative input).
  - 🔍 **Learning** from experience and **inference**.
- 🚀 Result: **Smarter, adaptable agents** that can handle new tasks efficiently.



Knowledge-Based Agent Cycle

👁️ 1. Perceive Environment

- The agent uses its **sensors** to gather percepts (signals, data, or observations) from the environment.
- These percepts provide **partial, possibly noisy information** about the current world state.
- 🧠 *Example:* A robot camera detects an obstacle; a chatbot reads a user's input message.

🧩 2. Update Knowledge Base (KB)

- The agent **integrates new percepts** into its **existing knowledge base**, refining its internal model of the world.
- This step may involve **removing outdated information** and **adding inferred facts**.
- 🗂️ *Purpose:* Keep the agent's knowledge **consistent and up to date** for accurate reasoning.

🧠 3. Reason & Infer

- The agent applies **logical inference rules** (e.g., Modus Ponens (طريقة الإثبات بالتأكيد), Resolution(طريقة الاختزال المنطقي)) to **deduce new knowledge**.
- It can infer **hidden properties**, **predict outcomes**, or **plan ahead** based on current facts.
- 💡 *Goal:* Move from "what is known" → "what logically follows."
- *Example:*
  - If `Rain(x) ⇒ Wet(x)` and `Rain(Street)`, then infer `Wet(Street)`.

⚙️ 4. Decide & Act

- The agent selects the **best or rational action** using the results of reasoning.
- Decision criteria often depend on a **performance measure** (utility, goal achievement, or reward).
- 🚀 *Example:* The robot chooses to turn left to avoid collision; the assistant replies with the most relevant answer.

🧠 Database vs. Knowledge Graph

🗃️ Database

- Stores **structured data** in tables (rows & columns).
- Focuses on **data storage and retrieval**.
- Relationships are **hard-coded** (foreign keys).
- Queries are **exact** (e.g., SQL).
- 💡 *Answers "What data do we have?"*

🕸️ Knowledge Graph (KG)

- Stores **entities and relationships** as a **graph (nodes & edges)**.
- Focuses on **meaning and connections** between data.
- Relationships are **first-class citizens** (easily queried).
- Supports **reasoning & inference**.
- 💡 *Answers "What does this data mean and how is it related?"*

🧠 Knowledge Base vs. Database — What's the Difference?

🔍 Aspect	🗃️ **Database**	🧩 **Knowledge Base**
Purpose	Store and retrieve raw data	Store, **organize**, and **reason** with knowledge
Data Type	Structured data (numbers, strings, records)	**Facts, rules, relationships, and logic**
Queries	SQL-like: "Give me all customers from Paris."	Logical inference: "Who lives in Paris and speaks French?"
Inference	❌ None – just retrieval	✅ Yes – can **derive new facts** using reasoning
Representation	Tables and rows	Logical sentences, semantic networks, ontologies
Example Use	Banking system, inventory	Expert system, search engine, intelligent assistant

🔧 When a Database Can Be a Knowledge Base

A database transcends its basic function and becomes a knowledge base when it incorporates two critical elements:

Semantic Richness

It stores not just raw data, but also the intrinsic meaning, relationships, and logical connections between data points.

Reasoning Engine Integration

It's coupled with an inference layer or reasoning engine capable of drawing logical conclusions and new insights from the stored information.

✅ Example: Hospital System

Database: Raw Data

```
Patient(Name="Ali", Temp=39, Cough=True)
```

```
Disease(Name="Flu", Symptom="Cough", Symptom="Fever")
```

Stores factual records without inherent intelligence for deduction.

Knowledge Base: Data + Rules

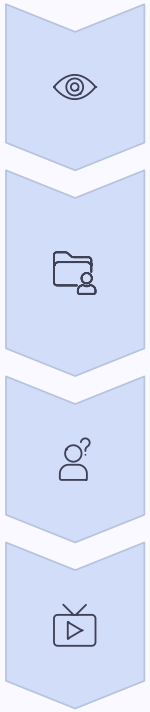
$$\forall x \, (\text{HasSymptom}(x, \text{Cough}) \land \text{HasSymptom}(x, \text{Fever}) \Rightarrow \text{HasDisease}(x, \text{Flu}))$$

By combining data with this rule, the system can infer that Ali likely has the flu, a capability a simple database lacks.

- A knowledge base keeps track of things
- We can TELL it facts or ASK for inference
- For example:
  - TELL: Father of John is Bob
  - TELL: Jane is John's sister
  - TELL: John's Father is the same as John's sister's father
  - ASK: Who's Jane's Father?

Knowledge-Based Agent Operations

A **knowledge-based agent** continuously interacts with its environment through a **sense-reason-act** cycle.
It uses a **knowledge base (KB)** and a **reasoning engine** to choose intelligent actions based on what it knows and perceives.



👁️ Perceive

→ Gather **new information** from the environment through sensors.

🧩 Tell

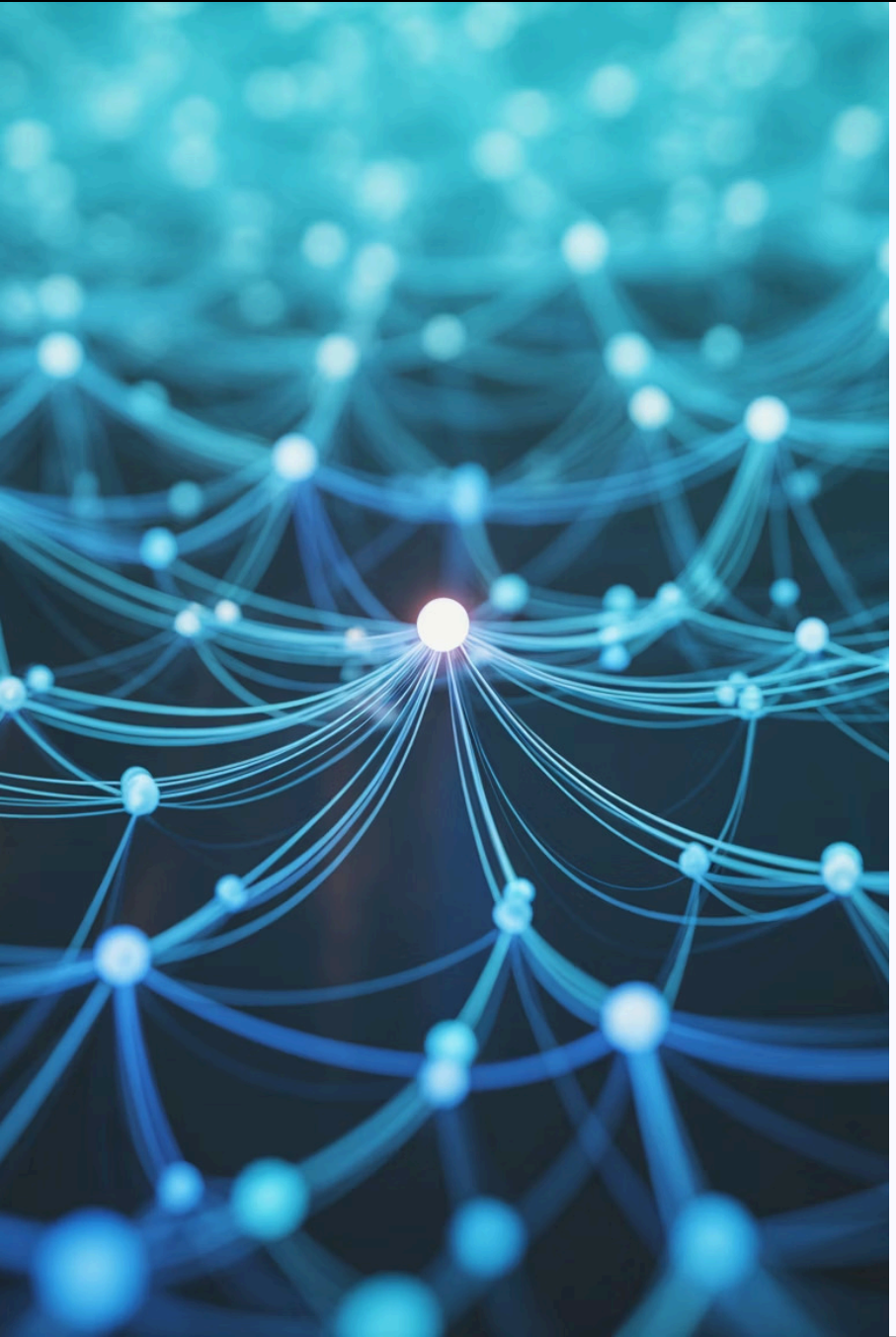
→ **Add** new percepts to the **Knowledge Base (KB)** to update internal knowledge.

❓ Ask

→ **Query** the KB using the **reasoning engine** to determine the best action.

⚙️Execute

→ **Perform** the chosen action using actuators, then observe the outcome.



Fundamental Concepts

1

🧠 Knowledge Base

- Repository of **facts and rules** about the world.
- Acts as the agent's **memory** and **foundation for reasoning**.
- Continuously updated as the agent perceives new information.

2

💬 Knowledge Representation Language (KRL)

- A **formal language** used to express knowledge logically.
- Must be **expressive** enough for complex relationships yet **efficient** for reasoning.
- Examples: **Propositional Logic**, **First-Order Logic (FOL)**.

3

🔍 Inference

- The process of **deriving new facts** from existing knowledge.
- **Sound inference:** never produces false conclusions.
- **Complete inference:** can derive **all truths** implied by the KB.

4

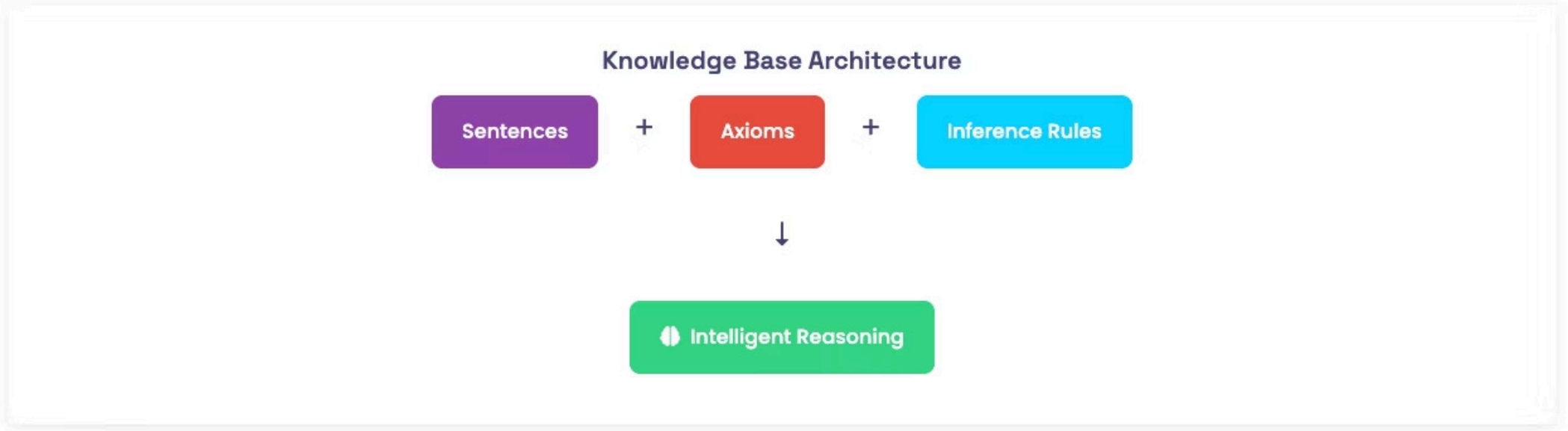
🌎 Background Knowledge

- General, pre-existing knowledge about **how the world works**.
- Helps the agent **interpret new situations** using known principles.
- Example: "If it rains, the ground becomes wet."

Anatomy of a Knowledge Base

💡 Knowledge Base Components

A knowledge base is more than just a storage system—it's a structured collection of **sentences**, **axioms**, and **inference rules** that enable intelligent reasoning.



Sentences: Facts & Rules

Sentences are the basic building blocks of a knowledge base. They represent facts about the world and rules that govern relationships between facts.

Facts (Atomic Sentences):

- Basic statements about the world
- Can be true or false
- Represent specific instances
- Examples: "It is raining", "John is tall"

Rules (Complex Sentences):

- If-then relationships
- General principles
- Enable inference
- Examples: "If it rains, then the ground is wet"

</> Example Sentences:

```
// Facts
is_bird(tweety)
has_wings(tweety)
is_red(apple1)
is_sweet(apple1)

// Rules
can_fly(X) :- is_bird(X), has_wings(X)
is_fruit(X) :- is_red(X), is_sweet(X)
if can_fly(X) then is_animal(X)
```

Axioms: Fundamental Truths

Axioms are fundamental truths that are assumed to be true without proof. They form the foundation upon which all other knowledge is built.

Logical Axioms:

- Basic logical principles
- Always true by definition
- Example: A ∨ ¬A (law of excluded middle)

Domain Axioms:

- Specific to the problem domain
- Assumed to be true
- Example: "All birds have wings"

Common Sense Axioms:

- General world knowledge
- Widely accepted truths
- Example: "Objects fall when dropped"

💡 **Example Axioms:**

```
// Logical Axioms
∀x (P(x) ∨ ¬P(x))                    // Law of excluded middle
∀x (P(x) → P(x))                     // Reflexivity
∀x,y (P(x) ∧ P(y) → P(x) ∧ P(y))     // Conjunction

// Domain Axioms
∀x (bird(x) → has_wings(x))          // All birds have wings
∀x (mammal(x) → warm_blooded(x))     // All mammals are warm-blooded
∀x (fish(x) → lives_in_water(x))     // All fish live in water
```

Inference Rules: Deriving New Facts

Inference rules specify how to derive new sentences from existing ones. They are the "engine" that powers logical reasoning.

Common Inference Rules:

- **Modus Ponens:** $P \rightarrow Q, P \vdash Q$
- **Modus Tollens:** $P \rightarrow Q, \neg Q \vdash \neg P$
- **Resolution:** $P \vee Q, \neg P \vee R \vdash Q \vee R$
- **Conjunction:** $P, Q \vdash P \wedge Q$

Properties:

- **Soundness:** Only derive true conclusions
- **Completeness:** Can derive all true conclusions
- **Efficiency:** Fast execution
- **Clarity:** Easy to understand and verify

Example Inference:

```
// Given knowledge base:
1. bird(tweety)                    // Fact
2. ∀x (bird(x) → can_fly(x))      // Rule
3. ∀x (can_fly(x) → is_animal(x)) // Rule

// Using Modus Ponens:
4. can_fly(tweety)                 // From 1, 2
5. is_animal(tweety)               // From 4, 3
```

🧠 Anatomy of a Knowledge Base

🧩 Structure & Components

- A **Knowledge Base (KB)** is a collection of **sentences** written in a **formal knowledge representation language**.
- Each sentence is an **assertion about the world** – something that can be **true** or **false** depending on the situation.
- **Axioms** are **fundamental truths or assumptions** accepted without proof.
  → They form the **foundation for all reasoning** within the KB

💡 In Short:

> A **Knowledge Base** = structured logical memory where **truths, rules, and axioms** work together to enable **reasoning and inference**.

⚙️ Key Properties

- 🧱 **Formal Syntax & Semantics** – every sentence follows strict logical structure and meaning.
- 💬 **Explicit Claims** – each statement clearly expresses a fact or rule.
- 🌍 **Truth-Dependent** – sentence truth values depend on the current **state of the world**.
- 🧭 **Axioms = Core Principles** – base knowledge used for inference.
- 🧠 **Compositional Structure** – complex knowledge is built from **simpler logical statements**.

🤝 Interacting with the Knowledge Base

🗣️ Tell Operation

- Adds **new knowledge** (sentences) to the KB.
- Used when the agent **perceives** something or **learns** a new fact.
- 🧩 *Example:* After sensing smoke, the agent tells: `Smoke(Room1)` → updates KB.

❓ Ask Operation

- Queries the KB to **retrieve or infer** information.
- The agent **asks** what actions or conclusions follow from its current knowledge.
- 💬 *Example:* "Is there fire in Room1?" → KB checks inference: $Smoke(Room1) \land Heat(Room1) \Rightarrow Fire(Room1)$.

🧠 The Declarative Approach

- Focuses on **telling the agent what is true**, not **how to act**.
- The agent uses **inference** to determine appropriate actions on its own.

💡 *Key Idea:*

> Instead of programming behaviors, we **describe knowledge**,
> and the agent **reasons** about what to do.

Two Levels of Agent Analysis

🧠 Knowledge Level vs. Implementation Level

📘 Knowledge Level

- Describes **what the agent knows** – independent of how it's represented.
- Focuses on the **content** and **logical relationships** of knowledge.
- Abstract view: *"What does the agent know and believe?"*
- 💡 Example:

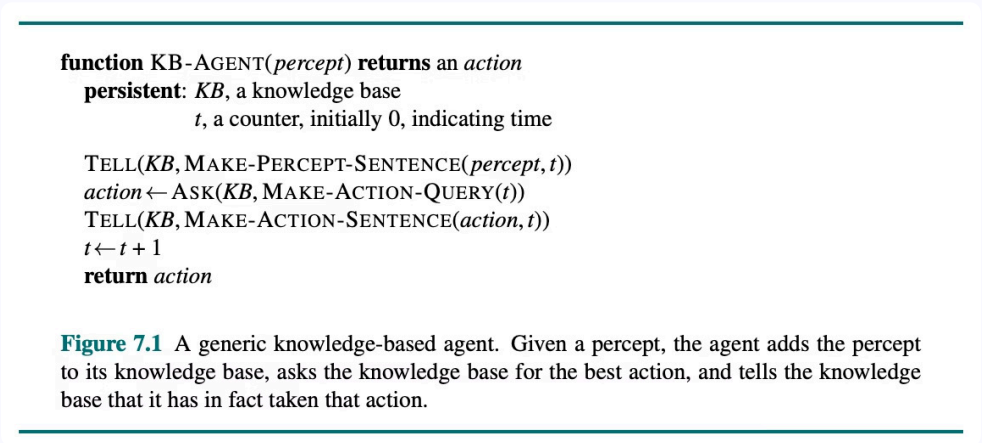
> The agent knows "If it rains, the ground becomes wet."
> (We don't care yet how this rule is stored.)

⚙️ Implementation Level

- Describes **how** the agent's knowledge is **represented and processed**.
- Focuses on **data structures**, **storage methods**, and **inference algorithms**.
- Considers **efficiency**, **scalability**, and **computational complexity**.
- 💡 Example:

> The rule is stored as a logical sentence or a data structure in memory
> and evaluated by an inference engine.

🤖 A Simple Knowledge-Based Agent



function KB-AGENT(*percept*) **returns** an *action*
  persistent: *KB*, a knowledge base
    t, a counter, initially 0, indicating time

  TELL(*KB*, MAKE-PERCEPT-SENTENCE(*percept*, *t*))
  action ← ASK(*KB*, MAKE-ACTION-QUERY(*t*))
  TELL(*KB*, MAKE-ACTION-SENTENCE(*action*, *t*))
  $t \leftarrow t + 1$
  return *action*

Figure 7.1 A generic knowledge-based agent. Given a percept, the agent adds the percept to its knowledge base, asks the knowledge base for the best action, and tells the knowledge base that it has in fact taken that action.

- 📚 **Maintains a Knowledge Base (KB)**
  - Stores facts and rules about the world.
  - Updated as the agent perceives new information.
- 👁️ **Perception Processing**
  - Converts each **percept** into a **logical sentence**.
  - Adds it to the KB using the **Tell operation**.
- ❓ **Action Selection**
  - Uses the **Ask operation** to query the KB.
  - Infers the **best action** based on current knowledge.
- 🧩 **Action Execution**
  - Adds the chosen action to the KB (as a record of what it did).
  - Executes the action in the environment.
- 🔁 **Continuous Cycle**
  - Repeats at each time step:
    Perceive → Update KB → Infer → Act → Repeat

The Wumpus (Monster) World: A Classic AI Problem

PEAS Description

🎯 Performance Measure

- +1000 → Grab gold
- -1000 → Fall in pit / eaten by Wumpus (Monster)
- -10 → Shoot arrow
- -1 → Each action

🌍 Environment

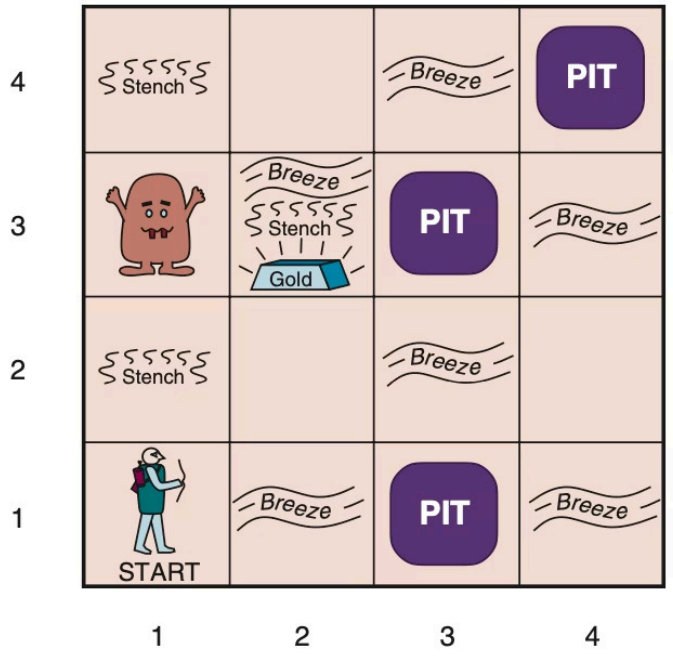
- Grid with pits, Wumpus, gold
- Adjacent to Wumpus → **Stench**
- Adjacent to pit → **Breeze**
- Gold → **Glitter** (shine/sparkle)
- One arrow, walls cause **Bump**, Wumpus death → **Scream**

👁️ Percepts (What Agent Senses)

- **Stench = Smell** (near Wumpus)
- **Breeze = Wind** (near pit)
- **Glitter = Shine** (gold present)
- **Bump = Hit wall**
- **Scream = Wumpus dies**

⚙️ Actions

- **Move:** Forward, Turn Left, Turn Right
- **Interact:** Grab, Release, Shoot



The Wumpus World provides a challenging testbed for knowledge-based agents. This grid-based environment contains hidden dangers and a single piece of gold.

Wumpus World Environment Characteristics

Understanding the properties of the Wumpus World helps us design effective agents. Let's analyze this environment using our standard taxonomy of environment types.

Partially Observable

The agent can only perceive adjacent squares, not the entire grid. This creates **uncertainty** about distant locations.

Deterministic

Each action has a **predictable** outcome. Moving forward always advances the agent one square if possible.

Sequential

Current decisions affect **future** options. The agent must plan sequences of actions to achieve goals.

Static

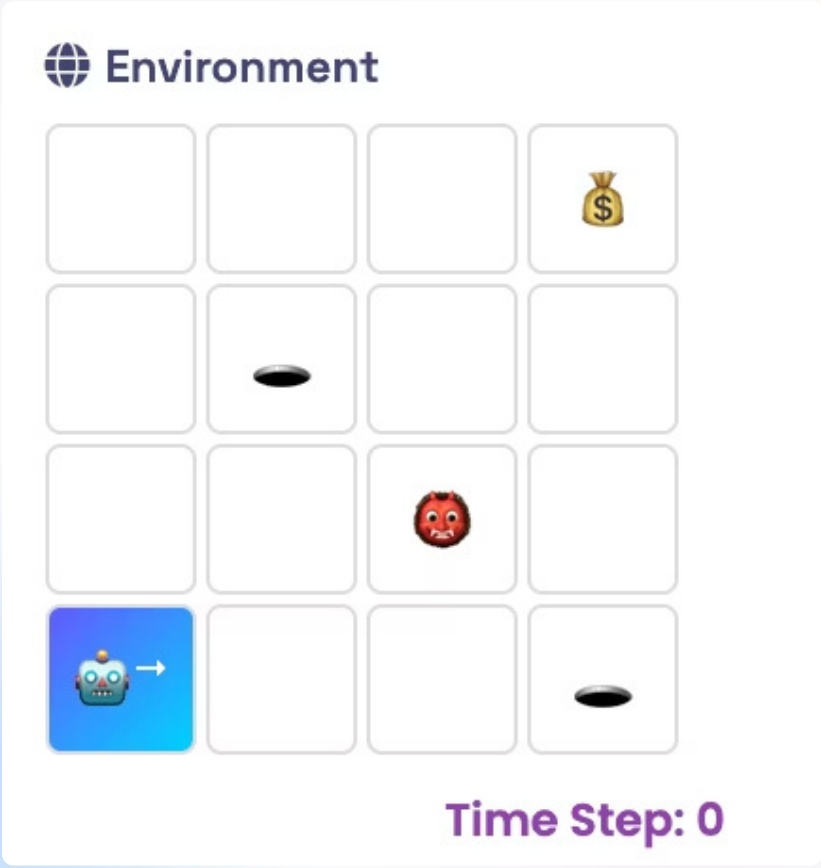
The Wumpus (Monster) and pits (holes) don't move. The environment only changes when the agent acts.

Discrete

Time advances in distinct steps. The agent's location and orientation are discrete values.

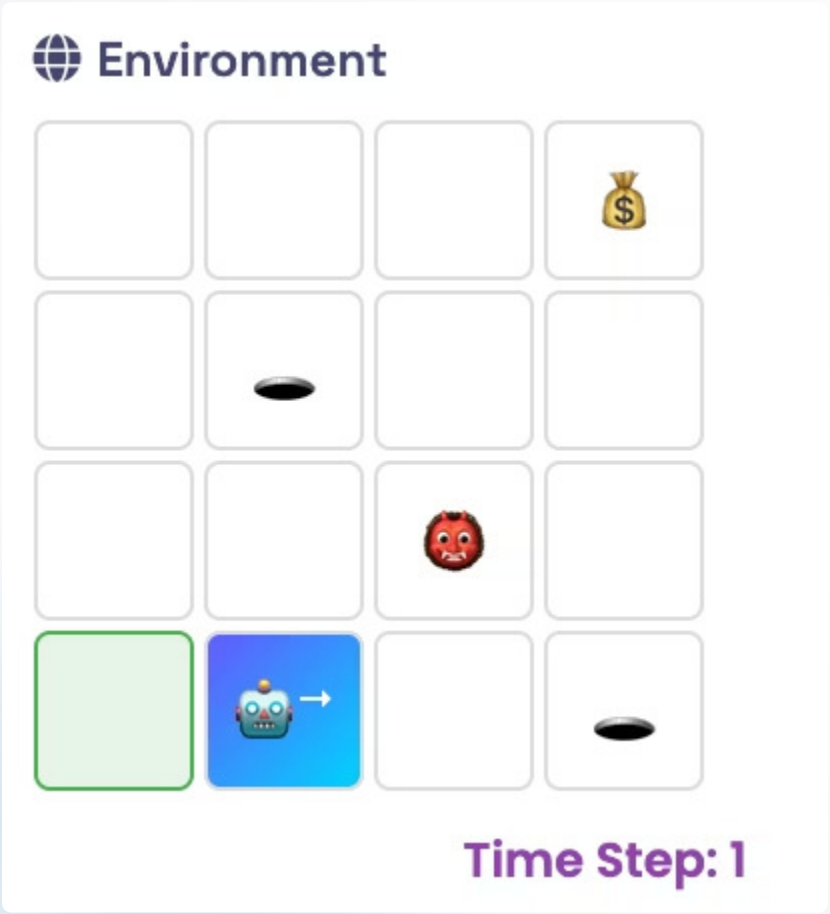
Single-Agent

The Wumpus is a natural hazard, not a competing agent with its own goals.



🏁 Beginning the Wumpus World

- 🧍 **Starting Point:**
  - Agent begins at **[1,1]** (bottom-left corner).
  - This square is **always safe** – no pits, no monster (Wumpus).
- 🌍 **Initial Knowledge:**
  - Agent knows only the **rules** of the world (how percepts relate to hazards).
  - It has **no prior information** about where gold or dangers are.
- 🧠 **Goal:**
  - **Find the gold** 💰 while **avoiding pits and the Wumpus** 👹.
  - Must **reason logically** using percepts (smell, breeze, glitter).
- 🔍 **Exploration Strategy:**
  - Move **cautiously** – infer safe squares before entering.
  - **Update the knowledge base** with each new percept.



👁️ Initial Observations – Wumpus World

- 🟩 **Starting Percepts at [1,1]**
- No **Stench** 🦨 → no Wumpus nearby
- No **Breeze** 🌬️ → no pits nearby
- No **Glitter** ✨ → no gold in current square

💡 *Even "nothing" is meaningful – absence of danger is knowledge!*

🧠 Agent's Inferences

- [1,1] is **safe** ✅ (no pit, no Wumpus)
- Adjacent squares **[1,2]** and **[2,1]** are **safe** (no breeze → no adjacent pits)
- The **gold is elsewhere** (no glitter)

🚶 Safe Moves

- Agent can safely move to **[1,2]** or **[2,1]**.
- Decisions are based on **logical inference**, not random exploration.

💡 Key Insight

> The agent uses **reasoning** – not **trial and error** –
> to explore the world **safely and intelligently**.

Exploring East: New Information

1

👁️ Percept at [2,1]

- 🌬️ **Breeze detected**
- 🚫 **No Stench** → no Wumpus nearby
- 🚫 **No Glitter** → no gold in this square

2

🧠 Immediate Inference

- The **breeze** means **at least one adjacent square** → [1,1], [2,2], or [3,1] **contains a pit** 🕳️

3

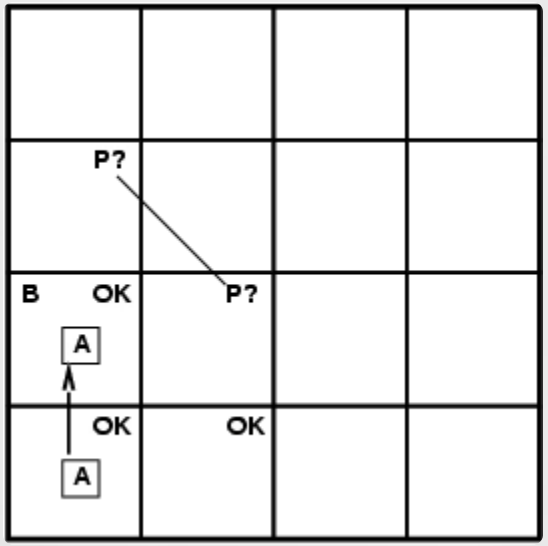
🔍 Combining Knowledge

- [1,1] is already known to be **safe** ✅
- Therefore, the **pit must be in** [2,2] **or** [3,1] **(or both)**

4

⚠️ Safety Assessment

- Moving **north to `[2,2]`** ❗ risky
- Moving **east to `[3,1]`** ❗ also risky
- Agent should **pause, reason further**, or **explore another direction**



⚠️ Reasoning About Danger

🧠 Current Situation

- The **breeze in `[2,1]`** signals a **pit nearby**, but the **exact location** is unknown.
- The agent must reason based on **partial knowledge**.

✅ What the Agent Knows

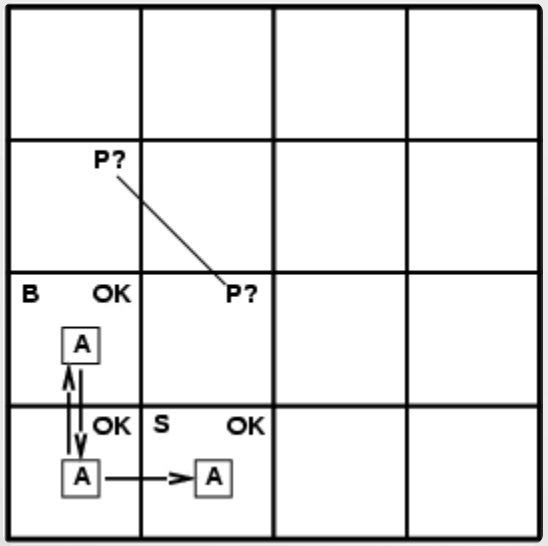
- `[1,1]`, `[2,1]`, `[1,2]` → **Safe squares** (visited / no breeze).
- At least one **pit** exists in `[2,2]` or `[3,1]`.
- **No Wumpus** in any visited or adjacent square.

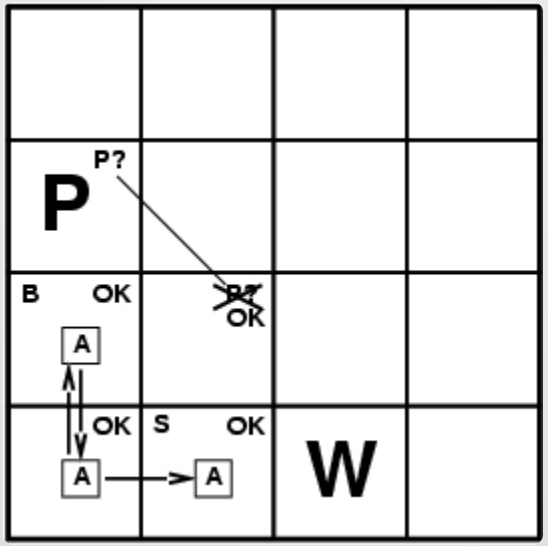
❓ What Remains Unknown

- Whether `[2,2]` has a pit.
- Whether `[3,1]` has a pit.
- Location of the **Wumpus** 👹.
- Location of the **Gold** 💰.

💡 Insight

> The agent operates under **uncertainty** –
> it must **gather more percepts** or use **logical / probabilistic reasoning**
> to decide the next safe move.





🧭 Strategic Exploration

🧠 Safe Decision-Making

- Instead of risking `[2,2]` or `[3,1]`, the agent **backtracks** to explore the **safe square `[1,2]`**.
- This **conservative strategy** prioritizes **safety** 🛡️ while still **gathering valuable information**.

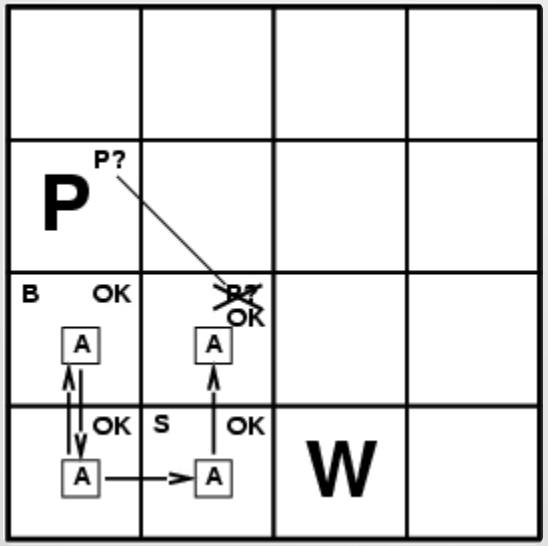
> 💬 Logical Reasoning in Action
>
> - The agent **infers** that exploring **safe squares first** is wiser.
> - Uses **logic**, not guesswork, to **avoid unnecessary risks**.
> - Demonstrates how **knowledge-based reasoning** guides exploration.

🗺️ Outcome

- By **systematically exploring safe areas**, the agent builds a **clearer map** of the world 🌍 before tackling **uncertain or risky regions**.

💡 Key Takeaway:

> Intelligent agents explore **strategically**, balancing **knowledge gain** with **risk avoidance**.



🧩Gathering Critical Information

👁️ New Percept at [1,2]

- 🦨 **Stench detected** → Wumpus is in an **adjacent square**: [1,1], [1,3], or [2,2].
  - [1,1] is already **safe** ✅
    → Therefore, **Wumpus is in** [1,3] **or** [2,2].

🌬️ Absence of Breeze

- No **Breeze** → **no pits** near [1,2].
- Thus [1,1], [1,3], and [2,2] are all **pit-free**.
- Combining earlier knowledge →
  The pit must be in [3,1] 🕳️

🧠 Logical Deduction

- Pit located: ✅ [3,1]
- Wumpus narrowed to: [1,3] or [2,2]
- Gold: still unknown 💰

💡 Key Insight

> Through **logical reasoning**, the agent transforms limited percepts into **precise knowledge** about the world –
> proving that **intelligence = reasoning, not guessing**.

✅ Definitive Conclusions

🕳️ Pit Location (Confirmed)

- **Square `[3,1]`** contains a **pit** – proven with certainty.

👹 Possible Wumpus Locations

- Wumpus must be in **`[1,3]`** *or* **`[2,2]`**.
- Both squares remain **dangerous and uncertain**.

🟩 Confirmed Safe Squares

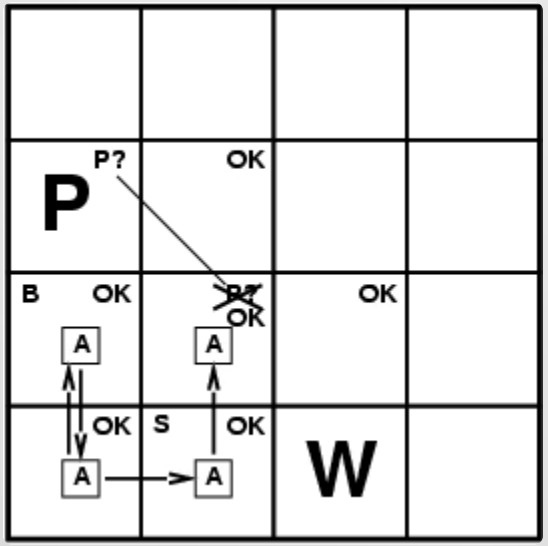
- **`[1,1]`**, **`[2,1]`**, and **`[1,2]`** are **completely safe** ✅.
- Agent can use these as **base positions** for further reasoning.

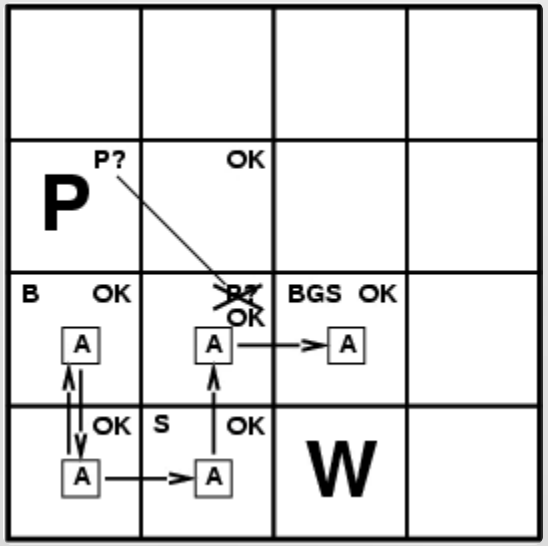
🧭 Next Steps

- Exploring **`[2,2]`** or **`[1,3]`** carries **equal risk** ⚠️.
- Agent must **gather new percepts** or apply **probabilistic reasoning** before proceeding safely.

💡 **Key Takeaway:**

> The agent combines multiple percepts to form **definite conclusions** – transforming **uncertainty into knowledge** through **logical inference**.





🎯 Optimal Strategy Emerging

🧠 Current Situation

- Agent has explored **[1,1]**, **[2,1]**, and **[1,2]**.
- The **knowledge base** now holds **accurate facts** about:
  - ✅ Safe squares
  - ⚠️ Possible Wumpus positions
  - 🕳️ Confirmed pit at **[3,1]** (must avoid).
- Goal: **Balance exploration with safety**.

🔍 Remaining Uncertainty

- **Wumpus** may be in **[1,3]** *or* **[2,2]** (equal probability).
- **Gold location** still unknown 💰.

🧭 Possible Next Moves

- ⚠️ **Enter [2,2]** → 50% chance of Wumpus.
- ⚠️ **Enter [1,3]** → 50% chance of Wumpus.
- 🏹 **Shoot arrow** → attempt to kill Wumpus from a safe position.
- 🚶 **Explore other safe directions** → gather more information first.

💡 **Strategic Insight:**

> The agent demonstrates **rational decision-making under uncertainty** – weighing **risk vs. knowledge gain** before acting.

Logic: The Foundation of Reasoning

💡 **Core Idea:** Logic is the **formal framework** that allows agents to represent knowledge and draw valid conclusions.



🧩 Syntax

- Defines **how to form valid sentences** in the logical language.
- Establishes the **grammar and symbols** for expressing facts and rules.
- Example:
- `A ∧ B` is **well-formed**, but `∧A B` is **not**.
- 💬 *Syntax = the form of expression.*



🌍 Semantics

- Specifies **what each sentence means** in relation to the world.
- Determines **when a sentence is true or false** in a given situation.
- Example:
- "It is raining" is true only in worlds where rain is actually falling.
- 💭 *Semantics = what the expression says about reality.*

🔗 Syntax + Semantics = Understanding

- **Syntax** builds the statements; **semantics** gives them **meaning**.
- A sentence alone is just a **string of symbols** – semantics tell us **what it implies about the world**.

Entailment



🧩 1. Intuitive Motivation

🔹 “What Does It Mean to ‘Know’ Something?”

Question

> “If the agent knows A, and from A we can logically conclude B,
> can we say the agent *knows* B too?”

Answer:
✅ Yes – **if** B *follows necessarily* from A.
That’s what **entailment** means.

💬 Simple example:

> - The agent knows: “If it rains, the ground gets wet.”
> - The agent also knows: “It’s raining.”
>   Then it *must* be true that “The ground gets wet.”

That “must” = **entailment**.

Definition of Entailment

> A sentence **α** is *entailed* by a knowledge base **KB**
> if **α must be true whenever KB is true.**

Symbolically (preview only):

> $KB \vDash \alpha KB \vDash \alpha$KB□α

💬 You can explain:

> “Entailment means that the truth of α is guaranteed by the truth of KB – it’s impossible for KB to be true and α to be false.”

Entailment (الاستلزام): When Truth Follows

- **Entailment** describes a **logical relationship** between statements –
  where the **truth of one** guarantees the **truth of another**.
- Formally:

> If the **Knowledge Base (KB)** is **true**,
> then every sentence **entailed by KB** must also be **true**.

- In other words, **α logically follows from KB** –
  there is **no situation** where KB is true and α is false.

Key Properties

- 🌍 **Semantic Relationship**
  Entailment concerns **truth**, not the structure or form of sentences.
  It depends on **meaning**, not syntax.
- ✅ **Truth Preservation**
  If **KB** ⊨ **α**, then **α can never be false** when **KB is true**.
- ⚙️ **Proof-Independent**
  Entailment does **not depend on how** you prove α –
  only on whether α is **true in all models** of KB.
- 🔒 **Necessary Logical Consequence**
  α is a **must-be-true** result of KB –
  its truth is **logically guaranteed** by the knowledge base.

Understanding entailment is crucial because it defines what an agent can legitimately conclude from its knowledge. An inference procedure is correct only if it derives sentences that are actually entailed by the KB.

Entailment (الاستلزام): When Truth Follows

💡 Core Idea

- **Entailment** (الاستلزام) describes when **one statement logically follows** from another.
- If the **Knowledge Base (KB)** is true, the **entailed sentence** must also be true.

Formal Definition

- We write **KB ⊨ α** to mean that the **Knowledge Base (KB)** *entails* the sentence **α**.
- This means: **α is true in every possible world where KB is true.**
- In other words, **the truth of KB guarantees the truth of α** – there is **no situation** where KB is true and α is false.

Example

- If

> **KB = {x = 2}**,
> then
> **KB ⊨ x + 2 = 4**,
> because whenever **x = 2**, it must logically follow that **x + 2 = 4**.

Understanding entailment is crucial because it defines what an agent can legitimately conclude from its knowledge. An inference procedure is correct only if it derives sentences that are actually entailed by the KB.



What Is a Model?

A **model** is one *possible world* –
a complete description of which facts are true or false.

💬 Example:
If our propositions are:

P: It's raining
Q: The ground is wet

Then each possible **model** is a combination of truth values for P and Q.

➡️ Each row = a *model*, i.e., one possible state of the world.

What Is a Model?

Then each possible **model** is a combination of truth values for P and Q.

Model	P	Q	Description
m_1	T	T	Raining & wet
m_2	T	F	Raining & dry
m_3	F	T	Not raining but wet
m_4	F	F	Not raining & dry

➡️ Each row = a *model*, i.e., one possible state of the world.



Truth in a Model

> A sentence is **true in a model** if it correctly describes that world. Otherwise, it's **false** in that model.

Example:
In model m_1 (raining & wet):

- "It's raining" → True
- "Ground is wet" → True
- "If it's raining, the ground is wet" → True

In model m_2 (raining & dry):

- "If it's raining, the ground is wet" → False

💬 Explain that models define what it *means* for something to be tru

Entailment and Models

> The Knowledge Base (KB) **entails** a sentence α
> if **every model** that makes KB true
> also makes α true.

In words:

> There is **no possible world** where KB is true and α is false.

Symbolically (optional preview):

$KB \vDash \alpha KB \vDash \alpha$KB⊨α

💬 Example:

- KB: {“It’s raining”, “If it rains, the ground is wet”}
- α: “The ground is wet”
- Every model that makes KB true also makes α true → ✅ entailment holds.



Why Models Matter to Agents

Concept	In Human Logic	In Agent Terms
Model	Possible world	Environment configuration
KB	What's known	Agent's knowledge
Entailment	What must be true	What the agent can safely conclude

💬 Example:

> If every possible cave layout (model) consistent with the KB has no pit in [1,2], then the agent can **safely move** there.

That's how **truth in models → safe action**.

🌍 Models and Entailment

💡 What Is a Model?

- A **model** is a **formally defined world** – a specific way the world *could be* – (a possible version of the world) that tells us **which sentences are true or false**.
- Logicians use models to **evaluate truth**:

> A sentence is *true in a model* if what it describes actually happens in that world.

📘 **Example:**
If a model m says "it is raining," then the sentence **Rainy** is *true in m*, but **Sunny** is *false in m*.

🧱 Model Definitions

- We say **m is a model of α** → if **α is true in m**.
- **M(α)** = the **set of all models** where α is true.

> (Think of it as *all worlds that make α true*.)

🔗 Entailment via Models

We define **entailment** formally using models:

KB α if and only if M(KB) ⊆ M(α)

📘 **Meaning:**

> Every world where the **Knowledge Base (KB)** is true is also a world where **α** is true.

In other words:

- The **truth-worlds of KB** are **contained inside** the **truth-worlds of α**.
- There is **no model** where KB is true and α is false.

Models and Entailment

🔗 Entailment via Models

We define **entailment** formally using models:

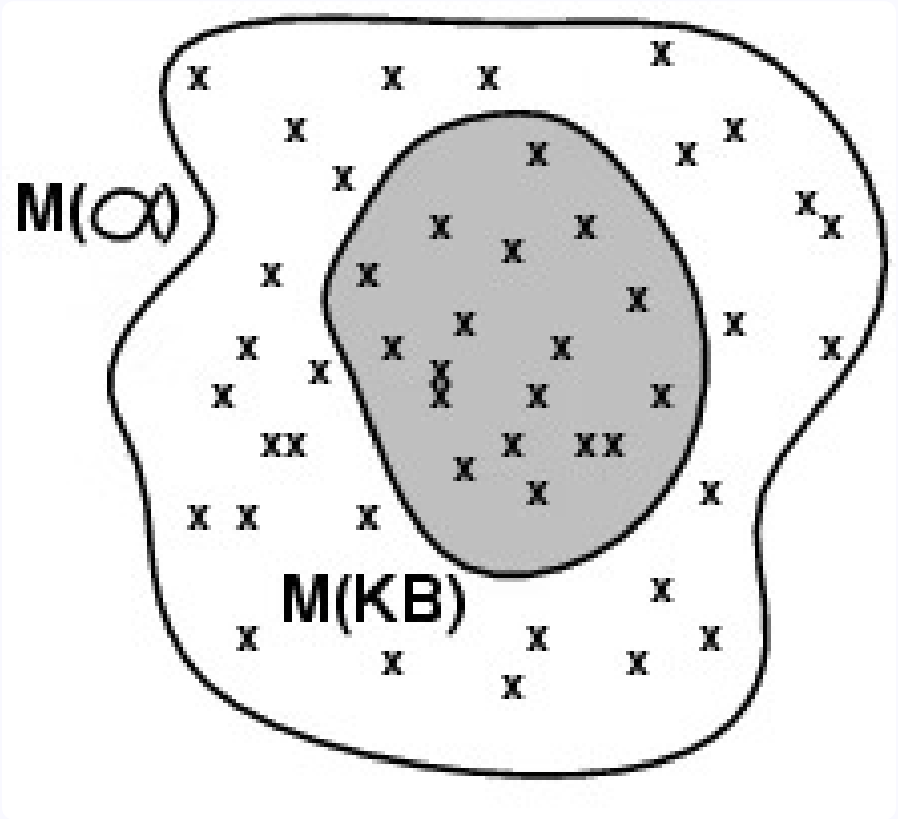
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📘 Meaning:

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Example: Entailment via Models

⚙️ Given Knowledge Base (KB):

KB = { It is raining → The ground is wet,
It is raining }

🎯 Claim (α):

α = The ground is wet

✅ Check entailment:

- **KB ⊨ α** means:
  In *every model* where KB is **true**, α is also **true**.
- From the table:
  - When KB is **true**, α is also **true** in all cases ✅.
  - There is **no world** where KB is true and α is false.

💡 Conclusion

> Therefore, **KB ⊨ α**
> ("The ground is wet" is *entailed* by the knowledge base).

🧩 Now let's see the models (possible worlds):

Model (world)	It is raining	Ground is wet	KB true?	α true?
1️⃣	True	True	✅ (both sentences true)	✅
2️⃣	True	False	❌ (violates "if raining → wet")	❌
3️⃣	False	True	✅ (KB is true because antecedent is false)	✅
4️⃣	False	False	✅ (KB is true because antecedent is false)	❌

Inference

We've established the semantic foundations of logic through entailment and models. Now we'll explore the computational side: how can we actually derive entailed sentences using algorithms? This leads us to inference procedures, logical equivalence, and important properties like validity and satisfiability.



🧩 1. Definition of Inference (Intuitive, Not Formal Yet)

🔹 "What Is Inference?"

> Inference is the process of **deriving new sentences (facts)** from existing knowledge.

Example:

```
If it's raining, the ground is wet.
It's raining.
Therefore, the ground is wet.
```

💬 Explain:

- The agent uses **logical reasoning** to discover what must be true.
- Inference produces **new knowledge** that wasn't explicitly told to the agent.

🧠 2. Why Agents Need Inference

- “Why Inference Matters”

Reason	Explanation
🧭 **Decision-making**	The agent can choose safe or optimal actions.
🧩 **Knowledge expansion**	It can deduce facts that weren’t directly observed.
🔍 **Partial observability**	Helps the agent reason about unseen parts of the environment.
💬 **Communication**	Allows compact representation of general knowledge (“If there’s a breeze, a pit is nearby”).

Example (Wumpus World):

> Agent perceives *no breeze* at (1,1).
> From rule: “If there is a breeze, there’s a pit nearby,”
> the agent can infer that adjacent squares (1,2) and (2,1) are **safe**.

⚙️ 3. The Inference Engine Concept

🔹 “The Inference Engine”

Show this diagram:

```
Percepts → Tell (add to KB)
KB + Inference Engine → Ask (query)
→ New Knowledge → Action Decision
```

Explain:

- The **Knowledge Base (KB)** holds what the agent *knows*.
- The **Inference Engine** applies **reasoning rules** to draw conclusions.
- Together, they form the agent’s “mind.”

💬 Analogy:

> KB = memory (facts)
> Inference Engine = thinking process

🧮 5. Simple Inference Examples

- “Examples of Reasoning in Action”

Percept	Rule	Inferred Knowledge
No breeze at (1,1)	If breeze → pit nearby	No pit in (1,2), (2,1)
Stench at (1,2)	If stench → Wumpus nearby	Wumpus in (1,3) or (2,2)
Glitter at (2,2)	If glitter → gold here	Safe to grab gold

This shows students *how logic looks like in action*.

🧩 6. Mention Inference Types (Conceptually)

- 🔹 Key Slide: “Types of Inference (Preview)”

Type	Direction	Idea
Forward Chaining	Data-driven	Start from known facts and derive consequences.
Backward Chaining	Goal-driven	Start from a goal and find supporting facts.
Model Checking (later)	Truth-driven	Check all possible worlds where KB holds true.

🧠 Say:

> “Next time, we’ll formalize how to perform these reasoning steps using propositional logic.”